

Will Wands

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EDUCATION

Rutgers University Honors School of Engineering

New Brunswick, NJ

B.S. Electrical & Computer Engineering and Computer Science — GPA: 3.72

Expected May 2029

- **Relevant Coursework:** Data Structures, Data Management, Linear Algebra, Probability, Discrete Mathematics

EXPERIENCE

The MITRE Corporation

Bedford, MA

Software Engineering Intern

May 2026 – Aug 2026

- Decreased translation retrieval latency by 30% by engineering multiple high-throughput RESTful API endpoints backed by rigorous unit test suites to ingest streaming data from a Redis priority queue.
- Automated real-time data transfers across high-compliance systems by architecting an on-demand microservice translation pipeline utilizing Vue.js, Python, and the BlackSheep asynchronous API framework.
- Delivered production-ready software features within a secure CI/CD pipeline by collaborating inside a 10-person Agile engineering team using Git-based peer code reviews.

NJ Governor's School of Engineering & Technology

New Brunswick, NJ

Research Scholar

Jul 2024 – Aug 2024

- Optimized path-planning vectors against real-world telemetry constraints by evaluating A* and RRT* graph algorithms inside a custom Python and Pygame testing framework.
- Validated autonomous navigation simulation models by co-authoring a technical research paper presented at the peer-reviewed MIT Undergraduate Research Technology Conference (URTC).

PROJECTS

Teamfight Tactics Teacher

C++, Python, REST API, TCP, Redis

May 2026

- Eliminated network bottlenecks and enforced strict API rate limits by architecting a C++ backend data ingestion pipeline using a custom Redis-style in-memory key-value store.
- Optimized analytical throughput by engineering an asynchronous engine that processes 50+ statistical fields per match via lambda-driven evaluation, exposing computed metrics over a JSON/HTTP socket interface to a real-time Python dashboard.

RUOnCampus – Off-Campus Housing Marketplace

React, Express.js, Supabase, PostgreSQL, Tailwind CSS

Jan 2026 – Present

- Accelerated search query speeds to under 40ms by mapping transactional relational schemas across 8 database tables.
- Reduced user onboarding drop-offs by 25% by writing a unified, multi-step listing composer interface utilizing React Context state stores and debounced lookup logic.
- Spearheaded a full-stack sublet matchmaking platform by deploying an Express.js backend backed by granular, token-based authentication workflows.

Angry Signing Llama – Real-Time CV Pipeline

Python, FastAPI, TensorFlow, MediaPipe, JavaScript

Feb 2026

- Attained a 97% sign recognition accuracy across 35 unique signs by training convolutional neural networks in TensorFlow and extracting structural body nodes via MediaPipe.
- Engineered sub-second inference windows by establishing low-latency web sockets between computer vision microservices and an optimized JavaScript frontend.

Imposture – Smart Posture Monitor

Python, OpenCV, MediaPipe, Arduino, C++

Mar 2026

- Achieved real-time physical posture correction alerts by architecting a computer-vision monitoring system that processes streaming video frames to detect ergonomic slouching thresholds.
- Eliminated communication bottlenecks between systems by writing a synchronous serial interface in C++ to transmit structural coordinate flags from MediaPipe to an external Arduino hardware network.

LEADERSHIP

IEEE Intelligent Ground Vehicle Competition

Rutgers University

Lead Programmer

Sep 2025 – Present

- Mitigated navigation errors by designing autonomous path-finding and localized spatial mapping systems utilizing OpenCV edge-detection logic to detect physical obstacle fields.
- Streamlined sensor integration across complex hardware systems by deploying custom object perception and LiDAR tracking pipelines using ROS 2 and Gazebo in Linux environments.

Chinese Student Organization

Rutgers University

Treasurer

Sep 2025 – Present

- Managed \$20,000+ worth of complex operational metrics, budget planning, and financial allocations from Rutgers University.
- Executed high-capacity community and charity initiatives, managing on-site logistics and registration pipelines to deliver seamless operations for 400+ attendees at events year round.

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL, JavaScript, HTML/CSS

Data Structures & Concepts: Multithreading, Asynchronous I/O, REST APIs, Graph Algorithms, Relational Indexing

Frameworks & Libraries: FastAPI, Express.js, React, Vue.js, BlackSheep, OpenCV, MediaPipe, TensorFlow

Databases & Infrastructure: Redis, PostgreSQL, Supabase, Docker, Kubernetes, Linux, Git, AWS